Chapter 13

The Mean Value Theorem and its Applications

§ Section: 4.2-4.3 MVT

13.1 The Mean Value Theorem

Recall that if $f(x)$ is differentiable at $x = a$, then (**in-the-small**)

$$f(x) \approx f(a) + f'(a)(x - a)$$

$$\Rightarrow f(x) - f(a) \approx f'(a)(x - a)$$

The Mean Value Theorem says that (**in-the-large**)

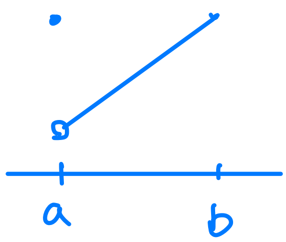
$$f(x) - f(a) = f'(c)(x - a) \textbf{ for some } c \textbf{ between } x \textbf{ and } a.$$

There is a lemma preparing for the Mean Value Theorem.

Rolle's Theorem

Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).
3. $f(a) = f(b)$



Then there is a number c in (a, b) such that $f'(c) = 0$.

Example 1.1

Proof of Rolle's Theorem:

Pf: By hypo ①, $f(x)$ attains global maxi & mini on $[a,b]$.

Let the global maxi be M. " mini " m. $\} \Rightarrow m \le f(a) \le M$.

$\boxed{\text{Case 1}}$ $m = f(a) = M$. $\Rightarrow f(x) = f(a)$ for $x \in [a,b]$.

Hence $f'(x) = 0$ on (a,b).

$\boxed{\text{Case 2}}$ If $m < f(a) = f(b)$, then m is a local mini and $m = f(c)$ for some $c \in (a,b)$.

By hypo ② $\Rightarrow f'(c)$ exists. Hence $f'(c) = 0$ by Fermat's Thm.

$\boxed{\text{Case 3}}$ If $M > f(a) = f(b)$, then M is a local maxi ...

Example 1.2

Prove that the equation $x^5 + 2x - 1 = 0$ has exactly one real root.

Sol: Let $f(x) = x^5 + 2x - 1$. $f(x)$ is conti on $\mathbb{R}$.

$f(0) = -1$, $f(1) = 2$. $f(0) < 0 < f(1)$.

By the IVT, there is some $c \in (0,1)$ s.t. $f(c) = 0$.

Suppose that there are $r_1 < r_2$ s.t. $f(r_1) = f(r_2) = 0$.

Then $f(x)$ is conti on $[r_1, r_2]$ and diff on (r_1, r_2).

By Rolle's Thm, there is some $d \in (r_1, r_2)$ s.t.

$f'(d) = 0 = 5d^4 + 2$ ✗ contradiction.

$\Rightarrow$ $f(x)$ has only one real root !

Example 1.3

Prove that $f(x) = x + 2e^x$ is 1-1.

sol: Suppose that there are $r_1 < r_2$ s.t. $f(r_1) = f(r_2)$.

$\because$ $f(x)$ is conti on $[r_1, r_2]$ and diff on (r_1, r_2),

$\therefore$ By Rolle's thm, there is some $c \in (r_1, r_2)$

s.t. $f'(c) = 0 = 1 + 2e^c$ ※ contradiction

$\Rightarrow$ $f(x)$ is 1-1.

The Mean Value Theorem

Let f be a function that satisfies the following hypotheses:

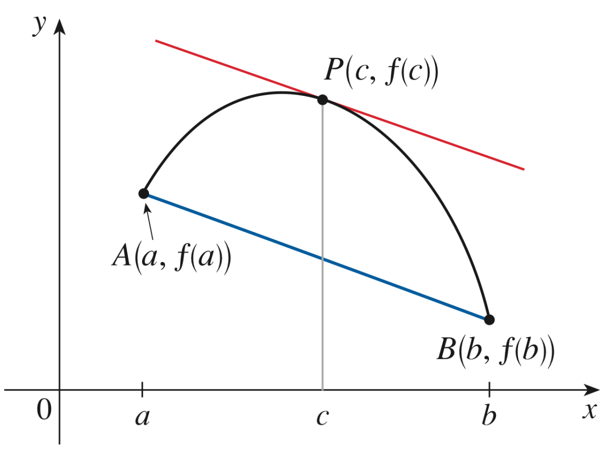
1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).

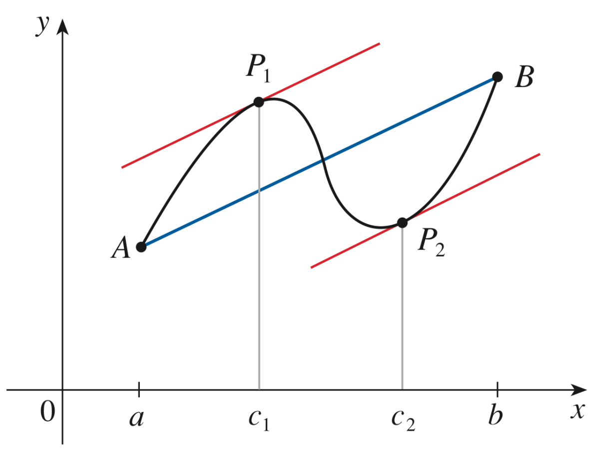
Then there is a number c in (a, b) such that

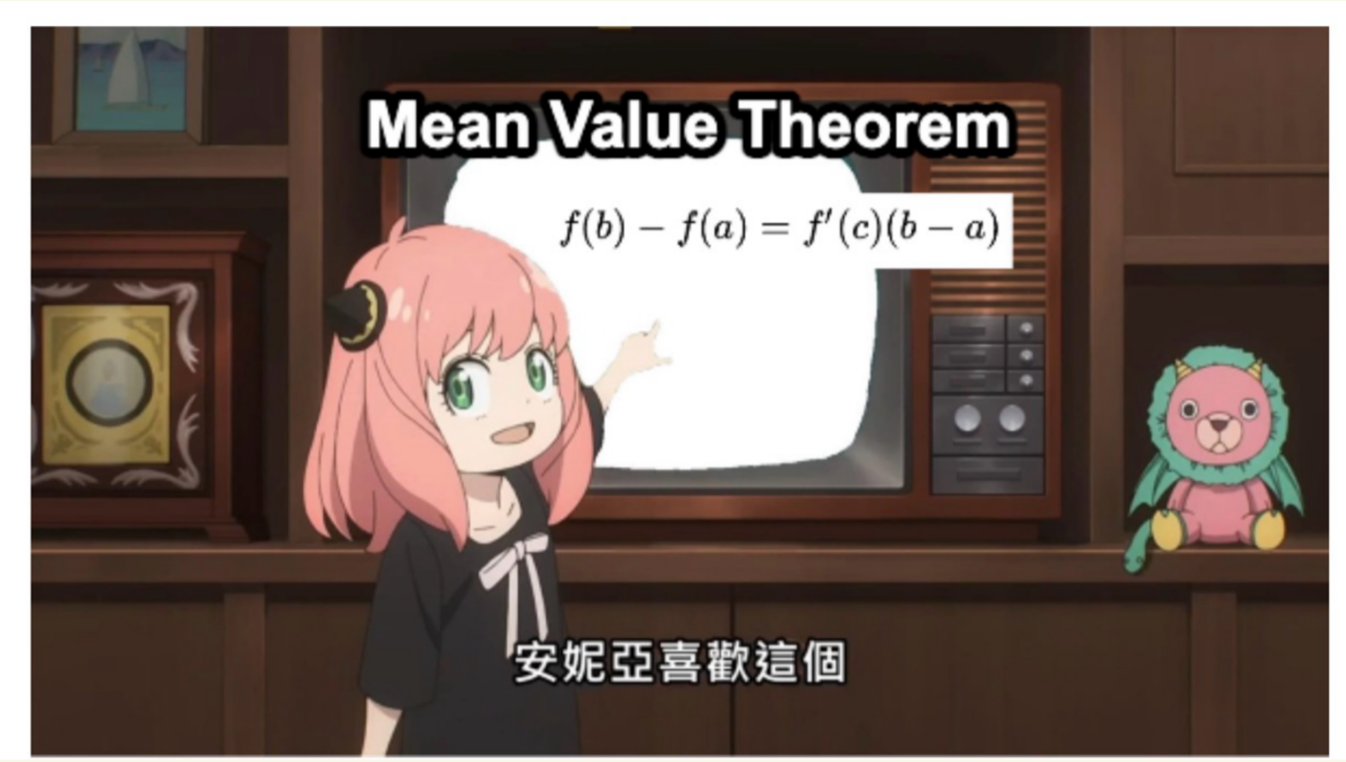
$$\boxed{f'(c) = \frac{f(b) - f(a)}{b - a}}$$

or, equivalently,

$$\boxed{f(b) - f(a) = f'(c)(b - a).}$$







希望 MVT 也是你的好朋友喔 !!

Example 1.4

Proof of The Mean Value Theorem.

Let $g(x) = f(x) - [f(a) + \frac{f(b)-f(a)}{b-a}(x-a)]$

① $g(x)$ is conti on $[a,b]$.

② $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, for $x \in (a,b)$

③ $g(a) = 0$, $g(b) = 0$

Rolle's Thm $\Rightarrow$ there is some $c \in (a,b)$ s.t.

$g'(c) = 0 = f'(c) - \frac{f(b)-f(a)}{b-a} \Rightarrow f'(c) = \frac{f(b)-f(a)}{b-a}$.

Applications:

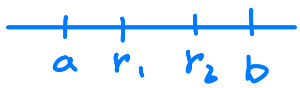
> **Theorem:**
>
> If $f'(x) = 0$ for all x in an interval (a,b), then f is constant on (a,b).

> **Corollary:**
>
> If $f'(x) = g'(x)$ for all x in an interval (a,b), then $f - g$ is constant on (a,b); that is, $f(x) = g(x) + c$ where c is a constant.

Example 1.5

Prove that if $f'(x) = 0$ for all $x \in (a,b)$. then f is constant on (a,b).

Sol: For any $r_1 < r_2$, $r_1, r_2 \in (a,b)$.



$[r_1, r_2] \subset (a,b)$.

f is conti on $[r_1, r_2]$ and diff on (r_1, r_2).

Thus $f(r_2) - f(r_1) \overset{MVT}{=\!=} f'(c)(r_2 - r_1)$ for some $c \in (r_1, r_2) \subset (a,b)$.

$= 0 \cdot (r_2 - r_1) = 0$

$\Rightarrow f(x)$ is a constant on (a,b).

Example 1.6

Simplify $\tan^{-1} x + \tan^{-1} \frac{1}{x}$.

Sol: Let $f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$ on $(-\infty, 0) \cup (0, \infty)$.

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\left(\frac{1}{x}\right)^2} \cdot \left(\frac{-1}{x^2}\right)$$

$$= \frac{1}{1+x^2} + \frac{-1}{1+x^2} = 0$$

$$f(x) = \begin{cases} c_1 & \text{on } (-\infty, 0) \\ c_2 & \text{on } (0, \infty) \end{cases},$$

Moreover,

$$f(-1) = \tan^{-1}(-1) + \tan^{-1}(-1) = -\frac{\pi}{2}$$

$$f(1) = \tan^{-1} 1 + \tan^{-1} 1 = \frac{\pi}{2}.$$

Hence

$$f(x) = \begin{cases} -\frac{\pi}{2}, & \text{for } x \in (-\infty, 0) \\ \frac{\pi}{2}, & \text{for } x \in (0, \infty) \end{cases}$$

Example 1.7

Show that $|\sin a - \sin b| \le |a - b|$.

Sol: Let $f(x) = \sin x$: diff on $\mathbb{R}$.

$$\sin a - \sin b = f(a) - f(b) \overset{\text{M.V.T.}}{=\!=} \underline{f'(c)(a-b)} = \cos c \cdot (a-b)$$

for some $c \in (a, b)$.

$$\Rightarrow |\sin a - \sin b| = |\cos c|\,|a-b| \le |a-b|$$

Example 1.8

Suppose that $f(0) = 2$, $f'(x) > 3$ for all x. Show that $f(x) > 2 + 3x$ for $x > 0$ and $f(x) < 2 + 3x$ for $x < 0$.

Sol: For $x > 0$, $f(x) - f(0) \overset{\text{M.V.T.}}{=\!=} f'(c) \cdot (x - 0) > 3 \cdot x$

for some $c \in (0, x)$. $\Rightarrow f(x) > f(0) + 3x = 2 + 3x$.

For $x < 0$,

> **The Racetrack Principle**
>
> Suppose that $f(0) = g(0)$ and $f'(x) > g'(x)$ for $x > 0$, then $f(x) > g(x)$ for $x > 0$.

Example 1.9

Proof of the Racetrack Principle

Sol: Let $h(x) = f(x) - g(x)$. $h(0) = f(0) - g(0) = 0$

$h'(x) = f'(x) - g'(x) > 0$ for $x > 0$.

For $x > 0$,

$h(x) = h(x) - h(0) \overset{\text{M.V.T.}}{=\!=} \underset{+}{h'(c)}\,\underset{+}{(x-0)} > 0$ for some $c \in (0, x)$.

$\Rightarrow h(x) > 0$ for $x > 0$

i.e. $f(x) > g(x)$ for $x > 0$.

Example 1.10

(a) Show that $e^x > 1 + x$ for $x > 0$.

(b) Show that $e^x > 1 + x + \dfrac{x^2}{2}$ for $x > 0$.

Sol: Let $f(x) = e^x - 1 - x$. f: diff. $f(0) = 0$.

$f'(x) = e^x - 1 > 0$ for $x > 0$.

For $x > 0$, $f(x) = f(x) - f(0) \overset{\text{M.V.T.}}{=\!=} \underset{+}{f'(c)} \cdot \underset{+}{(x-0)} > 0$ for some $c \in (0, x)$

$\Rightarrow f(x) > 0$ for $x > 0$

$\Rightarrow e^x > 1 + x$ for $x > 0$.

(b) Let $g(x) = e^x - 1 - x - \dfrac{x^2}{2}$. $g(0) = 0$

$g'(x) = e^x - 1 - x > 0$ for $x > 0$ by (a).

> **Theorem:**
>
> Suppose that $f(x)$ is continuous at $x = a$, and differentiable on an open interval containing a but possibly except a. If $\lim_{x \to a} f'(x) = L$, then $f(x)$ is differentiable at $x = a$ and $f'(a) = L$.

Example 1.11

Prove the above theorem.